

Animal models of human pregnancy and placentation: alternatives to the mouse

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Abstract

The mouse is often criticized as a model for pregnancy research as gestation is short, with much of organ development completed postnatally. There are also differences in the structure and physiology of the placenta between mouse and human. This review considers eight alternative models that recently have been proposed and two established ones that seem underutilized. A promising newcomer among rodents is the spiny mouse, which has a longer gestation than the mouse with organogenesis complete at birth. The guinea pig is also recommended both because it has well-developed neonates and because there is a wealth of information on pregnancy and placentation in the literature. Several smaller primates are considered. The mouse lemur has its advocates yet is less suited as a model for human pregnancy as its young are altricial, placentation very different from that of humans, and husbandry requirements not fully assessed. In contrast, the common marmoset, a New World monkey, has well-developed neonates and is kept at many primate centres. Marmoset placenta has some features that closely resemble human placentation, such as the interhaemal barrier, although it is uncertain if invasion of the uterine arteries occurs in this species. In conclusion, pregnancy research would benefit greatly from increased use of alternative models such as the spiny mouse and common marmoset.

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Introduction

Concern is often voiced about undue reliance on the mouse as a model for human health and disease. A news feature in *Nature* put it this way: ‘Mice have given a huge contribution to biology, but they can take it only so far’ (Roberts 2019). Both aspects of this statement resonate in relation to pregnancy research. On the one hand, the mouse has informed us about the earliest stages of embryonic development, a recent example being the application of *in vitro* systems to culture embryos of mouse and human (Bedzhov *et al.* 2014, Deglincerti *et al.* 2016, Shahbazi *et al.* 2016). On the other hand, the mouse has been soundly criticized as a model for the later events of human pregnancy and for placentation (Malassine *et al.* 2003, Schmidt *et al.* 2015). It is no coincidence that gene expression patterns in mouse and human placenta best agree when the comparison is restricted to the first 16 weeks of human pregnancy (Soncin *et al.* 2018).

This review speaks to whether there are realistic alternatives to the mouse as models for human pregnancy and placentation. A useful starting point was the news feature quoted previously, which listed several mammals that are closer to humans than any rodent (Roberts 2019). It highlighted the mouse lemur, which is a lower primate

from Madagascar. I shall consider these and other models that have been introduced or gained traction in recent years, such as the spiny mouse and chinchilla. Included are two animal models that have been underutilized of late, yet offer advantages compared to the mouse; these are the rabbit and guinea pig.

There are several questions we need to ask of any model. Do we have the genome? This usually is the case, although not all genomes are well annotated. Do we know how to keep the animals and breed them? Few scientists can afford to devote time to developing a model from scratch. Thus, a related question is whether the model is relevant in other areas of research, so that expertise in animal husbandry and breeding can be shared. Spiny mouse, for example, is of interest for wound healing (Pinheiro *et al.* 2018) and the naked mole-rat has been advanced as a model organism for ageing and pain research (Buffenstein 2005).

The aim of this review is to collate the information available on alternative animal models so that readers can make an informed judgement about their potential in a particular field of study. Relevant examples are given of previous research applications within pregnancy and placentation. However, no attempt is made to catalogue the topics or techniques for which a particular model is suited.

Reproductive strategies

One reason for choosing the mouse model was that mice are small, have a large litter size and short generation times (Schmidt *et al.* 2015). This reflects a reproductive strategy found in many mammals (*r*-selection) (Crews & Gerber 2003) but it is associated with the birth of poorly developed or altricial young that typically have closed eyes, no hair and are entirely reliant on parental care (Fig. 1A). Much of organ development takes place after birth, which makes the mouse a poor model for the later stages of human pregnancy when obstetrical syndromes such as preeclampsia and foetal growth restriction become manifest (Brosens *et al.* 2011). Other laboratory animals with altricial young include rats, hamsters and rabbits.

An alternative reproductive strategy (*K*-selection) results in precocial young that are well able to take care of themselves (Fig. 1B) (Crews & Gerber 2003). This requires a much longer gestation and smaller litters. Laboratory animals in this category include sheep,

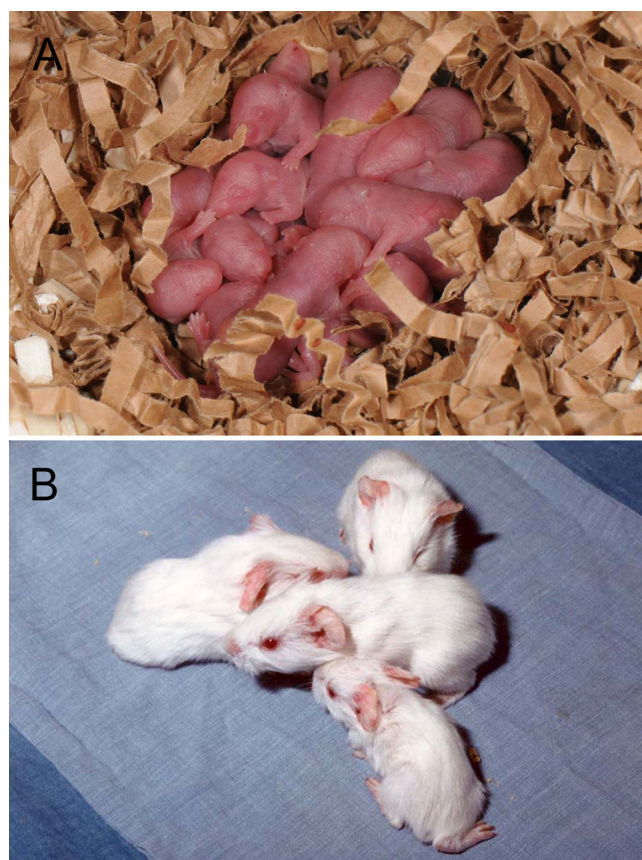


Figure 1 Altricial and precocial neonates are the result of different reproductive strategies placing different demands on the placenta. (A) Newborn litter of laboratory mice. The neonates have closed eyes and no hair and are entirely dependent on parental care (altricial). (B) One-day-old litter of guinea pigs. The neonates are relatively mature and mobile with open eyes and a full coat of hair (precocial). Photographs courtesy of Peter Bollen (A) and Per Svendsen (B).

guinea pigs and baboons. Human babies have little hair and are reliant on parental care, which led Adolf Portmann to characterize humans as secondarily altricial (Portmann 1941). This can be misconstrued, however, as a newborn baby is relatively well developed with open ears and eyes (Martin 2003). Organ development at birth is far advanced in humans compared to the mouse and other mammals with altricial young (Kurismaa 2020).

Evolutionary distance

The mammalian tree of life, as currently understood, is defined by molecular phylogenomics. Despite continued debate about the root of the tree, there is broad consensus on four major clades of eutherian mammal (Fig. 2A) (Murphy *et al.* 2007, Foley *et al.* 2016). Humans are found in Euarchontoglires, as are most of the model animals considered subsequently. It should be noted, however, that this clade has two major branches. One comprises primates, tree shrews and colugos (Euarchonta); the other rodents and lagomorphs (Glires).

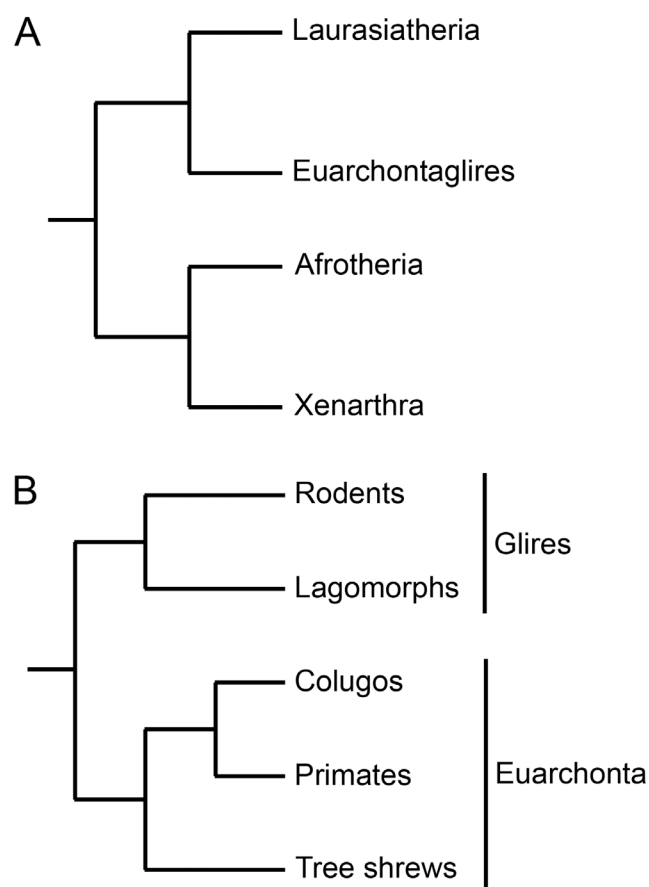


Figure 2 The mammalian tree. (A) The four major clades of eutherians (Murphy *et al.* 2007). (B) The orders of Euarchontoglires (Janecka *et al.* 2007). Note the separation of Glires (including rodents) from Euarchonta (including primates). There are alternative interpretations of the root of the tree and the position of tree shrews.

(Glires) (Fig. 2B) (Janecka *et al.* 2007). These two branches last shared a common ancestor in the Cretaceous period about 80 million years ago (Meredith *et al.* 2011).

Some model animals, such as the sheep, belong to a different clade (Laurasiatheria). Members of the two basal mammalian clades (Afrotheria and Xenarthra) are seldom proposed as models, although armadillos do have a villous placenta reminiscent of that in catarrhine primates (Enders 1960, 2002, Nelson *et al.* 1997).

Fetal membranes and placentation

Placentas come in a bewildering variety of shapes, internal morphologies and fetal–maternal interfaces (Mossman 1987). The primary function of the placenta is exchange of nutrients and gasses. Therefore, placentas are classified according to the cell layers separating maternal and foetal blood, the interhaemal barrier. Three main categories are recognized but each has many variants (Fig. 3). In an epitheliochorial placenta, the uterine epithelium remains intact and the interhaemal barrier comprises foetal capillary endothelium, one or more layers of trophoblast, uterine epithelium and maternal capillary endothelium (Fig. 3A). This type of placenta is often referred to as non-invasive but in ruminants binucleate trophoblast cells fuse with uterine epithelium to form trinucleate cells (Fig. 3B) or a hybrid syncytium (often referred to as synepitheliochorial). In an endotheliochorial placenta, the uterine epithelium is lost, and maternal capillaries are brought directly into contact with the trophoblast (Fig. 3C). In human placenta and most of the models subsequently considered, no maternal tissues are present in the interhaemal barrier. In these haemochorial placentas, the number of trophoblast layers varies from three in murine rodents (Fig. 3D) to one in hystricomorph rodents (Fig. 3E). A further layer of complexity is added by the arrangement of the two bloodstreams. In rodents and many other mammals exchange is between foetal and maternal capillaries. But in human placenta maternal blood perfuses the intervillous space, an arrangement found mainly in higher primates, but also seen in armadillos (Fig. 3F).

Despite this focus on the interhaemal barrier, there are important differences in the degree to which trophoblast cells invade the underlying endometrium and its blood vessels. Many obstetrical complications have their root in defective placentation resulting from the shallow invasion of the uterine spiral arteries by the trophoblast (Brosens *et al.* 2011). No animal model is fully suited to explore this aspect of human placentation (Carter 2011b).

Finally, early development of the human foetal membranes differs in many respects from mammals other than apes (Carter 2016, Ross & Boroviak 2020). Implantation is interstitial, resulting ultimately in a discoid placenta with a paraplacental structure incorporating elements of the decidua capsularis and

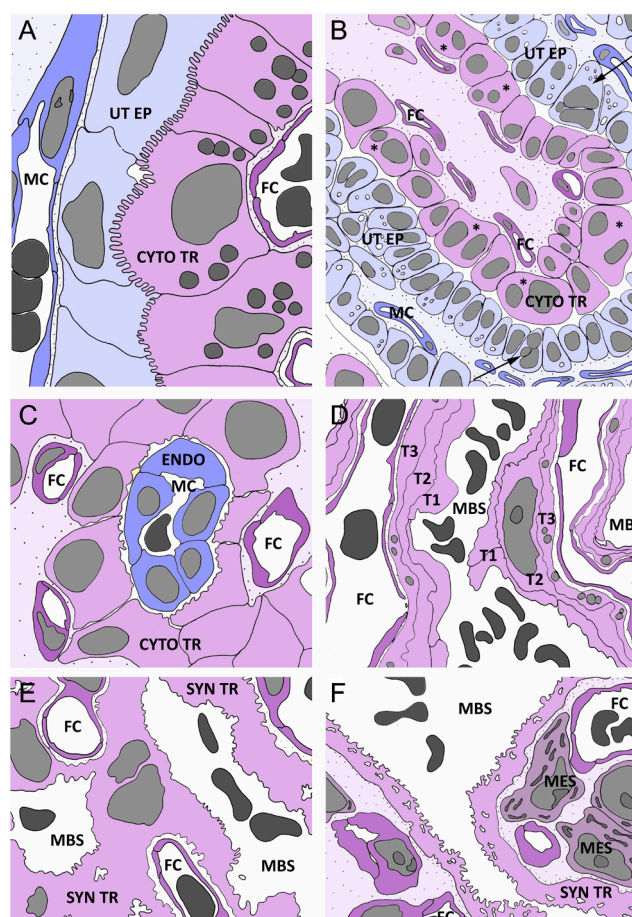


Figure 3 Interhaemal barrier of mammals. (A) Epitheliochorial placenta of a bush baby (*Otolemur crassicaudatus*). (B) Synepitheliochorial placenta of the cow (*Bos taurus*); note the binucleate trophoblast cells (asterisks) and trinucleate cells (arrows). (C) Endotheliochorial placenta of a bat (*Natalus* sp.); note the enlarged endothelial cells of the maternal capillary in contact with the trophoblast. (D) Haemotrichorial labyrinthine placenta of the mouse (*Mus musculus*). (E) Haemomonochorial labyrinthine placenta of the guinea pig (*Cavia porcellus*). (F) Haemomonochorial villous placenta of an armadillo (*Dasypus novemcinctus*). CYTO TR, cytotrophoblast; ENDO, endothelium; FC, foetal capillary; MC, maternal capillary; MBS, maternal blood space; MES, enlarged mesenchymal cell; SYN TR, syncytial trophoblast; T1, trophoblast layer 1 (cellular); T2, trophoblast layer 2 (syncytial); T3, trophoblast layer 3 (syncytial); UT EP, uterine epithelium. Based on material kindly supplied by Dr. Allen C. Enders, University of California at Davis, USA. Reprinted from Carter and Mess (2014) © 2014 with permission from Elsevier.

the amniochorion. The extraembryonic mesoderm appears very early and hinders contact between the yolk sac and maternal tissue, but a secondary yolk sac is found within the exocoelom and plays an important part in maternal–fetal exchange throughout the first trimester (Ross & Boroviak 2020). This feature is shared by all haplorrhine primates. Most other mammals conform to two other patterns of foetal membrane development (Carter 2016). In the first, the yolk sac

establishes contact with the uterine wall and forms a temporary yolk sac (choriovitelline) placenta. As a rule, the yolk sac placenta is short-lived and replaced by the chorioallantoic placenta. In lagomorphs, rodents and some other mammals, inversion of the germ layers, followed by loss of one wall of the yolk sac, results in an inverted yolk sac where the endoderm faces the uterine cavity. It plays an essential role in early embryonic development, continues to function after the formation of the chorioallantoic placenta and persists until term.

The placenta has functions other than maternal-fetal exchange, notably hormone secretion and interaction with the maternal immune system. These are reviewed elsewhere (Carter 2012) and only selected examples will be given here.

Large animal models

A few large mammals, such as sheep and Old World primates, continue to supply useful data and do not require a further endorsement. Information about these models should be sought elsewhere (Wolfe-Coote 2005, Barry & Anthony 2008, Carter 2011a).

The sheep (*Ovis aries*) remains an important model in foetal physiology since the ewe and foetus can be instrumented and studies performed after the acute effects of surgery and anaesthesia have subsided (Barry & Anthony 2008). There is the further advantage of a long gestation and precocial neonates. Placentation in sheep is cotyledonary and synepitheliochorial (Wooding & Burton 2008). Trophoblast does not invade the uterine vessels as it does in human pregnancy. In these respects, the sheep is inadequate as a model for human placentation.

Baboons (*Papio* spp.) and rhesus and cynomolgus macaques (*Macaca* spp.) are informative due to their closeness to humans. Their reproductive cycle resembles that of humans, with spontaneous ovulation and menstrual bleeding at the end of an infertile cycle (Martin 2007). Placentation in Old World monkeys and humans is alike in such respects as the structure of the villous trees, the presence of an intervillous space, and the transformation of the uterine spiral arteries (Carter et al. 2015). However, because colonies of large primates are expensive to maintain, they cannot be expected to replace the mouse.

There is an important body of literature concerning production animals such as cattle, swine and horses

that are studied for their own sake rather than as models of human pregnancy. Not infrequently, this research casts light on human development. A nice example is the expression of the *Oct4* gene by the trophectoderm lineage, where cattle resemble humans and rabbits while the gene is repressed in the mouse (Berg et al. 2011).

Rodents and lagomorphs

The orders Lagomorpha and Rodentia together constitute Glires (Douzery & Huchon 2004). Here I consider two species that have been underutilized in recent years and three that are relatively new (Tables 1 and 2).

Rabbit

For many years the European rabbit (*Oryctolagus cuniculus*) was the preferred model in embryology and reproductive biology. A recent review gives the historical context (Fischer et al. 2012). Notable examples are the work of Corner and Allen, which established the critical role of progesterone in pregnancy maintenance (Corner 1974), and Mossman’s description of countercurrent exchange in the placenta (Mossman 1926). For a while, rabbits went out of fashion, but there has been a welcome resurgence of interest (Fischer et al. 2012, Hue-Beauvais et al. 2019, Lopez-Tello et al. 2019). The rabbit is a good model to study foetal growth and foetal programming. Therefore the effects of maternal dietary restriction and other manipulations on foetal development have been explored in the rabbit (reviewed in Lopez-Tello et al. 2019). Rabbits have also been used to assess the impact of environmental pollutants on pregnancy and lactation (Hue-Beauvais et al. 2019). Disadvantages are that rabbits require a lot of space (Szendro & McNitt 2012) and that their neonates are altricial (Strauss 1957).

Rabbits have a discoid, labyrinthine and haemodichorial placenta. The interhaemal barrier includes two layers of trophoblast (Enders 1965). As described by Mossman, blood flows in the opposite direction in foetal capillaries and maternal blood channels, creating the anatomical basis for countercurrent exchange (Mossman 1926). This becomes important when the placental exchange is flow-dependent as is the case for oxygen (Carter 2015). The placenta rests on a thick bed of decidua within which the uterine arteries open into blood sinuses from which the placenta is supplied (Carter et al. 1971). The

Table 1 Lagomorphs and rodents (Glires) in use or proposed as alternative animal models to mouse and rat.

Species	Genome	Husbandry	Gestation	Neonate	Other applications
European rabbit	Yes	Well established	31 days	Altricial	Embryology and reproductive biology
Guinea pig	Yes (scaffold)	Well established	65 days	Precocial	Endocrinology of pregnancy
Long-tailed chinchilla	Yes (scaffold)	Bred for pelt	105–115 days	Precocial	Not widely used
Naked mole rat	Yes (scaffold)	Yes, but demanding	66–74 days	See text	Ageing research; cancer resistance
Cairo spiny mouse	Yes (scaffold)	Well described	38–39 days	Precocial	Wound healing; menstruation

Table 2 Placentation in lagomorphs and rodents (Glires) in use or proposed as alternative animal models to mouse and rat.

Species	Placental shape	Internal structure	Interhaemal barrier	Yolk sac placentation
European rabbit	Discoid	Labyrinthine	Haemodichorial	Inverted and lasts to term
Guinea pig	Discoid	Labyrinthine	Haemomonochorial	Inverted and lasts to term
Long-tailed chinchilla	Discoid	Labyrinthine	Haemomonochorial	Inverted and lasts to term
Naked mole rat*	Discoid	Labyrinthine	Haemomonochorial	Inverted and lasts to term
Cairo spiny mouse	Discoid	Labyrinthine	Haemotrichorial	Inverted and lasts to term

*Data not available; extrapolated from the dune mole-rat.

inverted yolk sac persists until term and this is the route by which maternal antibodies reach the foetus (Brambell *et al.* 1948).

Guinea pig

Guinea pigs (*Cavia porcellus*) were popular with a previous generation of scientists who used them to study placental circulation (e.g. Egund & Carter 1974, Herberger & Moll 1976, Peeters *et al.* 1982), placental gas exchange (Moll *et al.* 1970), maternal-fetal transfer (e.g. Reynolds & Young 1971, Schroder *et al.* 1975, Stulc & Svihovec 1977, Eaton *et al.* 1982, Peeters *et al.* 1986, van Dijk *et al.* 1986) and foetal growth restriction (Carter 1993). There is also an extensive body of work on the endocrinology of pregnancy (e.g. Blatchley & Donovan 1969, Challis *et al.* 1971). More recently, guinea pigs have been used to study the role of insulin-like growth factors and their receptors on placental transfer and foetal growth (Carter *et al.* 2005, Sferruzzi-Perri *et al.* 2007, Roberts *et al.* 2008). Current models of foetal growth restriction are based on progressive occlusion of the uterine artery (Lopez-Tello *et al.* 2018) or on maternal nutrient restriction (Roberts *et al.* 2001, Nevin *et al.* 2018, Maki *et al.* 2019). Guinea pigs need quite large cages and have a long gestation making them more

costly than mice. However, their neonates are precocial, which is a great advantage.

Guinea pig placenta is discoid, labyrinthine and haemomonochorial. In contrast to murid rodents, where the labyrinth and junctional zones form separate layers, the labyrinth of the guinea pig is disposed in lobes with interlobular areas supplied only by maternal vessels and corresponding to the murid junctional zone (Kaufmann & Davidoff 1977). The interhaemal barrier of guinea pig and other hystricomorph rodents comprises a single layer of syncytiotrophoblast (Fig. 4A) (Enders 1965). The arrangement of foetal capillaries and maternal blood vessels in the labyrinth favours countercurrent exchange (Miglino *et al.* 2004). A unique feature of hystricomorph placentation is the subplacenta from which trophoblast invades and transforms the uterine arteries (Mess *et al.* 2007). As in other rodents, an inverted yolk sac persists until term and is an important route for maternal-fetal transfer (King & Enders 1970).

Chinchilla

The long-tailed chinchilla (*Chinchilla lanigera*) has been proposed as an animal model as it has a small litter and a haemomonochorial placenta (Mikkelsen *et al.* 2017). The chinchilla is bred for its pelt and for a proof-of-concept

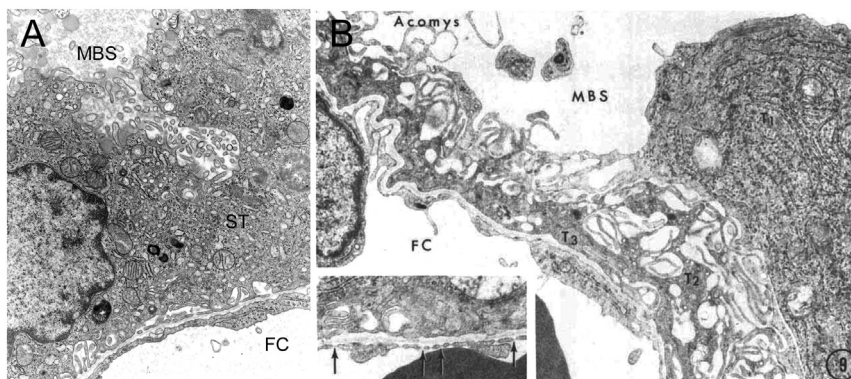


Figure 4 The exchange area of the placenta in two rodents. (A) The interhaemal area of the guinea pig (*Cavia porcellus*) is haemomonochorial. A single layer of syncytiotrophoblast (ST), its basal lamina and fetal endothelium separate the fetal capillary (FC) from the maternal blood space (MBS). Numerous microvilli project into the maternal blood space. Pockets containing microvilli are also found adjacent to the basal lamina. Courtesy of Dr Allen C. Enders. (B) The interhaemal membrane of the spiny mouse (*Acomys cahirinus*) is haemotrichorial. Trophoblast cells (T1) border the maternal blood space (MBS). The middle layer of syncytiotrophoblast (T2) shows complex folds on its outer surface. T3 is also syncytiotrophoblast. The nucleus (left) and cytoplasm of endothelial cells of the fetal capillary (FC) are shown, $\times 11,000$. The inset shows capillary endothelial fenestrations with diaphragms (arrows) $\times 19,200$. Reprinted with permission from King and Hastings (1977) © John Wiley & Sons Inc.

study, pregnant animals were obtained from dealers rather than raised in a laboratory (Mikkelsen *et al.* 2017, Overgaard *et al.* 2019). The foetus and placenta were imaged by CT and aspects of their metabolism explored by hyperpolarized MRI (Mikkelsen *et al.* 2017). In a second study, the distribution of ^{11}C -labelled metformin in the pregnant chinchilla was determined with PET; unexpectedly, metformin did not cross the placenta (Overgaard *et al.* 2019).

The placenta of the chinchilla is discoid, labyrinthine and haemomonochorial. As in the guinea pig, the labyrinth forms lobules separated by interlobular areas and there is a prominent subplacenta and an inverted yolk sac (Tibbitts & Hillemann 1959, King & Tibbitts 1976). The interhaemal barrier includes a single layer of syncytiotrophoblast (King & Tibbitts 1976). There is a rather narrow connection between the placenta and uterus termed the mesoplacenta (Tibbitts & Hillemann 1959).

Naked mole-rat

An unusual species promoted as an alternative animal model is the naked mole-rat (*Heterocephalus glaber*) (Buffenstein 2005, Roellig *et al.* 2011). This is an African hystricomorph with a unique life history. Colonies of naked mole-rats are extended family groups averaging 80 individuals with only one breeding female or queen (Jarvis & Sherman 2002). Gestation lasts about 70 days, average litter size is 11 pups and the queen delivers more than 50 pups a year (Jarvis & Sherman 2002). Foetal development is relatively slow (Roellig *et al.* 2011) and although pups can move around after birth, their eyes remain closed until 30 days postpartum. Therefore they do not fit the definition of precocial young (Roellig *et al.* 2011). Another aspect of the naked mole-rat's life history is longevity with individuals often living >20 years (Jarvis & Sherman 2002, Buffenstein 2005). They are, therefore employed as a model in ageing research (Buffenstein 2005, Buffenstein *et al.* 2020). They are also of interest due to their apparent resistance to cancer (Tian *et al.* 2013). Colonies of naked mole-rats can be kept in captivity (e.g. Roellig *et al.* 2011), but necessarily require special housing constructed of a series of chambers and tubes.

There is no description of placentation in this species, but the placenta of the dune mole-rat (*Bathyergus janetta*) is discoid, labyrinthine and haemomonochorial. It shows the characteristic features described previously for the guinea pig, including a subplacenta (Luckett & Mossman 1981).

Spiny mouse

The most promising newcomer is the Cairo spiny mouse (*Acomys cahirinus*), which is a murid rodent like the

mouse, rat or gerbil. Importantly, the spiny mouse has a longer gestation than the mouse and the neonates are precocial with open eyes and a coat of fur (Pinheiro *et al.* 2018). Organ development is also more advanced at birth. Nephrogenesis, for example, is complete (Dickinson *et al.* 2005). There is a model of foetal growth restriction in spiny mouse based on partial occlusion of the uterine artery by silastic tubing, which prevents the increase in blood flow that otherwise occurs with advancing gestation (Dickinson *et al.* 2017).

The endometrium of the spiny mouse undergoes spontaneous decidualisation during the oestrous cycle followed by shedding of tissue and blood at the end of an infertile cycle, that is, menstruation (Bellofiore *et al.* 2017, 2018). Indeed, a recent survey concluded that spiny mouse is the most promising model for studies of menstruation (Catalini & Fedder 2020). The spiny mouse is also a model for wound healing and tissue regeneration (Pinheiro *et al.* 2018, Maden & Brant 2019, Maden & Varholick 2020), so there is a growing community of scientists experienced in the husbandry of this rodent. One group has published a detailed account of the husbandry of the spiny mouse and recommends housing in standard rat cages (Dickinson *et al.* 2005, Dickinson & Walker 2007).

The placenta of the spiny mouse is discoid, labyrinthine and haemochorial (Assheton 1905). As in the mouse and other murid rodents, the interhaemal barrier comprises three layers of trophoblast (Fig. 4B) (King & Hastings 1977). There are sex-specific differences in development between the placentas of male and female fetuses (O'Connell *et al.* 2013). As in all rodents, the inverted yolk sac persists until term.

Primates and their kin

There are two suborders of primate initially defined by differences in the development of the foetal membranes (Hubrecht 1897) and more recently confirmed by phylogenomics (Murphy *et al.* 2007). Strepsirrhini includes the seven families of lemurs and lorises whilst Haplorrhini includes tarsiers, New World and Old World monkeys and apes (Fig. 5). The news feature on mouse lemurs mentioned several alternatives including tree shrews, strepsirrhine primates, tarsiers and marmosets (Tables 3 and 4) (Roberts 2019). Breeding colonies of these species have been maintained at one time or another, but continuity is found only for lemurs and marmosets, which are bred at several primate centres. Therefore, mouse lemurs and marmosets will be considered in greater detail than the other species.

Tree shrew

Tree shrews are closer to primates than rodents, although placed in a separate order (Scandentia) (Fig. 2B).

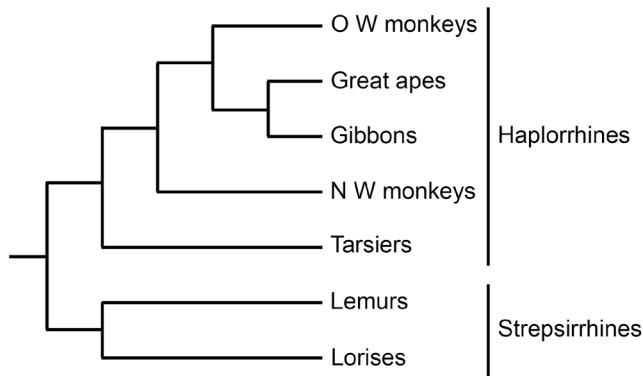


Figure 5 Classification of primates (Perelman *et al.* 2011).

Strepsirrhines and haplorrhines are suborders, tarsiers are regarded as an infraorder whilst the other clades shown are superfamilies. OW, Old World; NW, New World.

Their precise relation to colugos (Dermoptera) and primates is unclear (Janecka *et al.* 2007, Mason *et al.* 2016, Springer & Gatesy 2016). Tree shrews have a shorter gestation than most primates and the young are altricial, being born without hair and with eyes closed (Ferner *et al.* 2010). The placenta of the Northern tree shrew (*Tupaia glis*) is discoid, labyrinthine and endotheliochorial (Kaufmann *et al.* 1985, Luckhardt *et al.* 1985). A true yolk sac placenta persists at least to the early limb bud stage of development (Luckett 1968). Tree shrews have been bred in captivity, but the altricial neonate and endotheliochorial placenta argue against tree shrews as a good model for human pregnancy.

Mouse lemur

Mouse lemurs are strepsirrhine primates from Madagascar (Fig. 5). The species suggested as models are the grey mouse lemur (*Microcebus murinus*) and rufous mouse lemur (*M. rufus*), both of which are abundant in the wild. They are small, which is perceived as an advantage. Recent studies of these primates have been done in the field by capture, release and recapture of individual lemurs (Ezran *et al.* 2017). Breeding colonies of *M. murinus* have been kept on

Madagascar (Andriantsiferana *et al.* 1974) and elsewhere (Petter-Rousseaux 1962, Martin 1972, Zimmermann *et al.* 2016). However, a recent survey concluded that the husbandry requirements of mouse lemurs require further assessment (Lazaro *et al.* 2017). Gestation is short for a primate, about 60 days, and two to three young are born (Martin 1972, Andriantsiferana *et al.* 1974, Zimmermann *et al.* 2016). They are naked at birth with closed eyes and ears, that is, altricial (Petter-Rousseaux 1962, Jenkins 2003). There is a reference genome for *M. murinus* (Larsen *et al.* 2017) and current efforts focus on identifying naturally occurring null mutations (Ezran *et al.* 2017). Mouse lemurs mimic the cognitive ageing patterns described in humans and, therefore, are used as a model of brain disorders and Alzheimer's disease (Verdier & Mestre-Francés 2016).

The placenta of *M. murinus* and its development have been well described (Reng 1977, Strauss 1978a,b) from specimens collected 90 years ago by Bluntschli (Carter 2019). A yolk sac placenta is formed initially but soon displaced by the definitive chorioallantoic placenta (Reng 1977). The latter is discoid and labyrinthine. There is an extensive paraplacenta. Strepsirrhine primates (lorises and lemurs) tend to have epitheliochorial placentation (Carter *et al.* 2015). However, only the paraplacenta of the mouse lemur is epitheliochorial, there being loss of uterine epithelium in the central region, so that it is more accurately characterised as syndesmochorial (Reng 1977).

Bush baby

Demidoff's bush baby (*Galagoides demidovii*), like the mouse lemurs, belongs to the suborder of strepsirrhine primates (Perelman *et al.* 2011) (Fig. 5). Neonates of Demidoff's bush baby are precocial (Strahl 1899, Kingdon 1997). A yolk sac placenta supports early development. The chorioallantoic placenta of bush babies is diffuse and epitheliochorial and has chorionic vesicles to absorb uterine gland secretions (Strahl 1899, Gérard 1929, Njogu *et al.* 2006).

Table 3 Primates and kin (Euarchonta) in use or proposed as alternative animal models to catarrhine primates.

Species	Genome	Husbandry	Gestation	Neonate	Other applications
Northern tree shrew	No, but available for <i>Tupaia chinensis</i> (scaffold)	A breeding colony of <i>T. glis</i> has been kept	ca. 45 days	Altricial	Not widely used
Demidoff's bush baby	No, but available for <i>Otolemur garnetti</i> (scaffold)	Specialized centres, for example, Duke Lemur Center	110–114 days	Precocial	Not widely used
Grey mouse lemur	Yes	Several breeding colonies	ca. 60 days	Altricial	Alzheimer's disease and other brain disorders
Spectral tarsier	No, but available for <i>Carlito syrichta</i> (scaffold)	Breeding colonies difficult to maintain	ca. 193 days	Precocial	Not widely used
Common marmoset	Yes	Primate centres	ca. 144 days	Precocial; twins or triplets	Neuroscience; neurodevelopmental disorders

Table 4 Placentation in primates and kin (Euarchonta) in use or proposed as alternative animal models to catarrhine primates.

Species	Placental shape	Internal structure	Interhaemal barrier	Yolk sac placentation
Northern tree shrew	Discoid	Labyrinthine	Endotheliochorial	Temporary yolk sac placenta
Demidoff's bush baby	Diffuse		Epitheliochorial	Temporary yolk sac placenta
Grey mouse lemur	Discoid with extensive paraplacenta	Labyrinthine	Syndesmochorial	Temporary yolk sac placenta
Spectral tarsier	Discoid	Trabecular with intervillous space	Haemomonochorial	Secondary yolk sac
Common marmoset	Double discoid	Trabecular with intervillous space	Haemomonochorial	Secondary yolk sac

Tarsier

The closer affinity of tarsiers to monkeys and apes than to lemurs was first shown through studies of their foetal membranes (Hubrecht 1897). They belong to the suborder of haplorrhine primates though sufficiently distinct to be classed as an infraorder (Fig. 5) (Perelman *et al.* 2011). Gestation lasts about 193 days and there usually is a single precocial neonate (Roberts 1994, Gursky 2007). Attempts to maintain breeding colonies of tarsiers have been rather unsuccessful (Fitch-Snyder 2003). The spectral tarsier (*Tarsius spectrum*) has a discoid, haemomonochorial placenta (Hill 1932, Luckett 1974). The placental villi are joined by trabeculae giving an architecture similar to that described for the marmoset (Hill 1932, Enders & Carter 2012). As in other haplorrhines, early differentiation of the extraembryonic mesoderm leads to the formation of a secondary yolk sac confined to the exocoelom (Hubrecht 1896).

Marmoset

Marmosets and other New World monkeys are closer to the human lineage than lemurs or tarsiers, but not as close as Old World monkeys (Fig. 5). The common marmoset (*Callithrix jacchus*) is kept at many primate centres and its husbandry requirements are well known (Tardif *et al.* 2003). Because of a smaller body size, caging and feeding costs are lower than for macaques (Mansfield 2003, Riesche *et al.* 2018). Marmosets are used in several areas of biomedical research (Mansfield 2003) and are a particularly important model in neuroscience (Kishi & Okano 2017). Indeed, they are so sought-after that animals sometimes are in short supply (Servick 2018). Transgenic marmosets, most recently generated by CRISPR-Cas technology, are models of neurodevelopmental disorders (Kishi & Okano 2017, Yoshimatsu *et al.* 2019).

Gestation in marmosets lasts 143–144 days and the neonates are precocial (Tardif *et al.* 2003). Dizygotic twins are usual in free-living marmosets, but triplets are common in colonies kept at primate centres (Tardif *et al.* 2003). Unusually for mammals, the foetuses share a common placenta. It does not differ in mass between twin and triplet pregnancies (Rutherford & Tardif 2008). This is likely because placental growth plateaus around 100 days, whereas the acceleration in foetal growth comes later (Riesche *et al.* 2018). Thus, comparisons

between twin and triplet pregnancies in marmosets constitute a natural model of competition for shared resources (Rutherford & Tardif 2008). The placenta may be more efficient in triplet pregnancies (Rutherford & Tardif 2008) reflecting adaptations that include an increase in surface area at the fetal-maternal interface (Rutherford & Tardif 2009). As in other callitrichines (marmosets, tamarins and lion tamarins), competition between siblings continues after birth and not all survive to maturity (Tardif *et al.* 2003, McCoy *et al.* 2019). Under well-fed conditions, captive marmosets may become obese, which can impact on prenatal and later development making marmosets an ideal model to study the effects of obesity in pregnancy (Riesche *et al.* 2018). For example, a recent study looked at the effect of maternal weight on methylation of placental genes (Narapareddy *et al.* 2020). Such observations promise well for the use of marmosets to study the developmental origins of adult health and disease (Hanson & Gluckman 2014). An alternative approach is prenatal exposure to dexamethasone, a synthetic glucocorticoid, which has far-reaching effects on postnatal behavioural development (Hauser *et al.* 2008) and predisposes to metabolic syndrome (Nyirenda *et al.* 2009). In a separate application of the marmoset model, Zika virus was shown to cross marmoset placenta and affect foetal brain development (Seferovic *et al.* 2018).

Important work on preimplantation development in marmosets has shown greater affinity in gene expression of early cell lineages between marmoset and human than mouse (Boroviak *et al.* 2018). Further, it has been possible to compare common marmoset, rhesus macaque and human development with a particular focus on the secondary yolk sac (Ross & Boroviak 2020). As in all primates except gibbons and great apes, implantation in the marmoset is superficial with early penetration of the uterine epithelium and invasion of the uterine glands (Smith *et al.* 1987, Enders & Lopata 1999). Several authors have observed retrovirus particles around the time of implantation, presumably the products of endogenous retroviral genes (Smith & Moore 1988, Enders & Lopata 1999).

The mature marmoset placenta comprises two discs of about the same size. The placenta is villous rather than labyrinthine, but the villi are connected by bridges of trophoblast called trabeculae, giving a different architecture than in Old World monkeys and apes (Fig. 6) (Wislocki 1932). Early in pregnancy, the

interhaemal barrier includes syncytiotrophoblast and a contiguous layer of cytotrophoblast that corresponds to Langhans' layer in human placenta (Jollie 1973). Few cytotrophoblasts are found in late pregnancy and the interhaemal barrier then comprises syncytiotrophoblast with its basal lamina and the foetal capillary endothelium (Fig. 6A) (Jollie 1973, Wynn *et al.* 1975). Jollie, who studied the cotton-headed tamarin (*Saguinus oedipus*), remarked on the close similarity to human placenta and recommended callitrichines as a model for studies of placental transport (Jollie 1973).

The maternal blood supply of the marmoset placenta is rather unusual as there are intraplacental maternal capillaries from which blood reaches the intervillous or intertrabecular space (Fig. 6C) (Merker *et al.* 1987, 1988). We have suggested there is trophoblast invasion of the arteries (Fig. 6D) (Carter *et al.* 2015). This requires confirmation and the route of invasion needs to be established. Wynn observed isolated cytotrophoblasts within the decidua as well as long streamers of syncytiotrophoblast in contact with the endometrial vessels (Wynn *et al.* 1975). Another distinctive feature of the platyrrhine placenta is the occurrence of haemopoietic islets associated with the foetal capillaries (Fig. 6B) (Hill 1932, Wislocki 1943, Jollie *et al.* 1975, Merker *et al.* 1988).

Discussion

My foray into comparative placentation has shown me that much can be learnt from studying a wide range of species (Carter 2020). For experimental work, however, practical considerations limit one's choice. The mouse became popular because of the availability of embryonic stem cells, which offered an opportunity to create transgenic animals. An additional factor was the economy offered by their small size, short gestation and large litters. My aim is not to further criticise the mouse, but to explore if there are realistic alternatives for pregnancy research.

The main disadvantage of the mouse is that a brief gestation leads to the birth of altricial young. The same applies to many other murid rodents, such as the rat, and is also a drawback of the rabbit. The spiny mouse is a better model for pregnancy research since organogenesis is advanced at birth much as in human pregnancy (Dickinson *et al.* 2005). Hystricomorph rodents such as the guinea pig nearly all give birth to precocial young. Guinea pig remains the best of the classical models and there is a wealth of data on the physiology and endocrinology of pregnancy as well as foetal growth. The relatively large litters of guinea pigs reflect selective breeding over thousands of years (Rood & Weir 1970, Clutton-Brock 2012). Litter size is smaller in the chinchilla, which is advantageous in some settings, for example, imaging of the foetus and placenta (Mikkelsen *et al.* 2017). In naked mole-rats, the queen

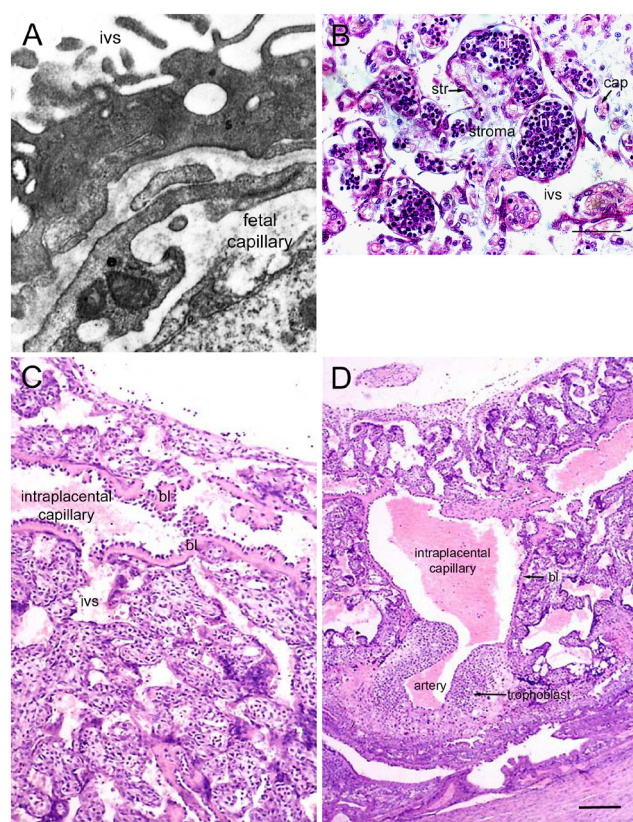


Figure 6 Placentation in marmosets. (A) Interhaemal membrane of cotton-headed tamarin (*Saguinus oedipus*) in late gestation (crown-rump length of twins 75 and 80 mm). During late pregnancy, the membrane is haemomonochorial, and the syncytiotrophoblast is supported by the subepithelial basal lamina, $\times 25,200$. Reprinted with permission from Jollie (1973) © John Wiley & Sons Inc. (B) Placenta of the common marmoset (*Callithrix jacchus*) showing prominent haemopoietic foci (hf). They are larger than a fetal capillary (cap). The villi are covered by a thin layer of syncytiotrophoblast (str); ivs, intervillous space. Scale bar 50 μ m. J. P. Hill Collection (Hapale P106) Museum für Naturkunde Berlin. Reprinted from Carter and Mess (2013) © 2013 Museum für Naturkunde Berlin. (C) Common marmoset placenta at 60 days of gestation. A maternal vessel with hypertrophied endothelium supported by a thick basal lamina (bl) has openings that bring maternal blood into contact with the syncytiotrophoblast of the developing villi. Scale bar 160 μ m. Boyd Collection, Centre for Trophoblast Research Cambridge. Reprinted from Enders and Carter (2012) © 2012 with permission from Elsevier. (D) Common marmoset placenta at 60 days of gestation showing transition between a uterine artery and an intraplacental capillary. The thick basal lamina beneath the capillary endothelium is clearly visible. No basal lamina is seen beneath the artery, which seems devoid of endothelium and is lined by a thick layer of cells suggested to be trophoblast. Scale bar 105 μ m. Boyd Collection, Centre for Trophoblast Research Cambridge. Reprinted from Carter *et al.* (2015) © 2015 The Authors, Published by The Royal Society.

delivers large litters of altricial young. Moreover, this species is demanding to keep. Though worth study in its own right (e.g. Blecher *et al.* 2020), it is not a realistic model for pregnancy research. It would, however, be

useful to know more about the foetal membranes and placenta of the naked mole-rat.

There is a deep divergence in the mammalian tree between rodents and primates (Fig. 2B). As highlighted by Roberts, mouse lemurs are closer to humans than any rodent (Roberts 2019). However, it is unclear if they are suitable laboratory animals. Much research has been done in the field on Madagascar (Ezran *et al.* 2017). Pregnancy research would need to be conducted at a dedicated primate centre. To date, there are no reports on interventions during pregnancy such as dietary restriction. A further disadvantage is that the neonates of mouse lemurs are altricial.

Marmosets have a suitable size and life history and are less costly to maintain than higher primates such as macaques and baboons (Mansfield 2003, Riesche *et al.* 2018). As detailed previously, they have already proved their worth for studies of the developmental origins of adult health and disease.

Whilst newer models being developed for foetal instrumentation is not anticipated, they could be amenable to foetal imaging as demonstrated for the chinchilla (Mikkelsen *et al.* 2017, Overgaard *et al.* 2019). They are also open to modern approaches such as transcriptomics (Carter 2018) and to gene editing by CRISPR-Cas and derived methodologies as already applied to marmosets (Yoshimatsu *et al.* 2019).

Placentation

Many of the major obstetrical syndromes are associated with defective placentation (Brosens *et al.* 2011). With that in mind haplorrhine primates are better models than strepsirrhine primates, rodents or rabbits. Lemurs and lorises have epitheliochorial or syndesmochorial placentation. In rodents and the rabbit, the placenta is discoid, labyrinthine and haemochorial. The number of trophoblast layers in the interhaemal barrier ranges from three (mouse, rat, spiny mouse) through two (rabbit) to one (guinea pig, chinchilla, mole-rat). Although it often is suggested that the haemomonochorial placenta makes for a better model than the haemotrichorial placenta (e.g. Mikkelsen *et al.* 2017), this has never been documented. More significant by comparison with human is the variation in how deeply trophoblast invades the uterine arteries. Invasion of decidual arteries is shallow in the mouse (Redline & Lu 1989), whereas in guinea pigs trophoblast reaches the level of the mesometrial arteries (Verkeste *et al.* 1998). In human pregnancy, foetal growth restriction and preeclampsia are associated with shallow invasion of the uterine spiral arteries. Conversely, in a rat model of preeclampsia, trophoblast invasion was deeper than usual (Geusens *et al.* 2010). Additionally, though uterine natural killer cells play a role in uterine artery transformation both in rodents and primates, they operate through completely different sets of receptors;

killer immunoglobulin-like receptors in primates (Carter 2011b, Penman *et al.* 2016) and lectin-like receptors (Ly49) in mice (Croy *et al.* 2003). Finally, rodents and lagomorphs have an inverted yolk sac with important transfer functions throughout gestation.

Placentas have important functions other than maternal–fetal transfer. Thus, placental hormones and cytokines affect maternal physiology, adapting it to the requirements of pregnancy and later lactation (Napso *et al.* 2018). The placenta of the mouse and perhaps that of other rodents secretes a variety of placental lactogens. They have their origin in duplication of the prolactin gene, whereas human placental lactogen is derived from the growth hormone gene (Soares 2004, Carter 2012). Two placental lactogens (PL-1 and PL-2) are important luteotrophic factors in murine rodents. In human pregnancy, this is a function of chorionic gonadotropin (hCG). The search for a chorionic gonadotropin in guinea pigs met with only partial success (Humphreys *et al.* 1982, Bambra *et al.* 1984).

Placentation is also an important consideration in choosing a primate model. Trophoblast is minimally invasive in mouse lemurs with only the uterine epithelium being lost, creating an interhaemal barrier that is syndesmochorial (Reng 1977). Moreover, the gross organisation of the placental disc is labyrinthine and in no way approaches the villous structure of Old World monkeys and apes. In the tarsier and marmoset, the interhaemal barrier at term has a single layer of trophoblast as in Old World monkeys and apes. Moreover, in marmoset, though the villi are connected by bridges of trophoblast (trabecula), there is clearly an intervillous or intertrabecular space. This seems also to be the case in tarsiers (Enders & Carter 2012). In contrast to the human, however, the intervillous space of the marmoset is supplied with maternal blood through openings in intraplacental maternal capillaries rather than from spiral arteries opening through a basal plate. A key question is whether the arteries supplying these capillaries are invaded by trophoblast and, if so, by what route (Carter *et al.* 2015).

Regarding placental endocrinology, the bush baby does not have a placental lactogen (Adkins *et al.* 2001) and this is likely true of the mouse lemur. It has been suggested that placental expression of the *GH* gene was present in the common ancestor of platyrrhine and catarrhine primates but has evolved along separate pathways (Papper *et al.* 2009). The marmoset has a cluster of eight genes and pseudogenes, including analogues of human placental lactogens (Wallis & Wallis 2006). As to chorionic gonadotropins, strepsirrhine primates and tarsiers have only the pituitary *LH* genes (Maston & Ruvolo 2002). Following the evolution of placental expression, the chorionic gonadotropin β -unit has continued to evolve. In the marmoset it has a rather different structure than hCG (Simula *et al.* 1995).

Conclusions

An alternative to the mouse ought preferably have a longer gestation and precocial young. This requirement is not met by the rabbit, naked mole-rat, tree shrew or mouse lemur. Apart from the rabbit, the husbandry requirements of these species either are not well-established or, in the case of the mole-rat, quite demanding. The spiny mouse and guinea pig do have precocial young and their husbandry requirements are known. Placentation in the spiny mouse resembles that of the mouse. The guinea pig has a single layer of trophoblast in the interhaemal barrier, although this difference from murid rodents should not be overemphasised. No rodent exhibits trophoblast invasion of uterine arteries that bears close comparison with human pregnancy.

Tarsiers are interesting but historically it has been difficult to maintain breeding colonies. Therefore, the most promising small primate is the common marmoset. It has a long gestation and gives birth to precocial twins or triplets. Colonies are maintained at many centres and the marmoset is an important model in fields such as neuroscience. The interhaemal barrier of marmoset placenta closely resembles that of human placenta. There are villi and an intervillous space though the overall structure is less open than in baboons, macaques and apes. Unlike in human placenta, there are intraplacental maternal capillaries. It remains to be established whether and how trophoblast invades the uterine arteries.

In conclusion, the future of pregnancy research is bright and will benefit greatly from alternative models not least the spiny mouse and common marmoset.

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Author contribution statement

A C conceived the topic, reviewed the literature and wrote the manuscript.

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